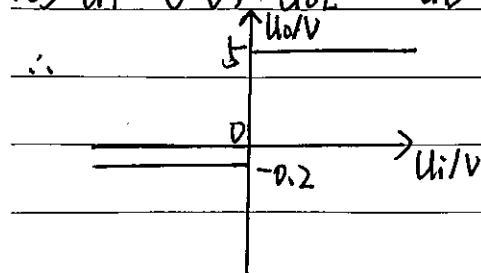


(b) $U_T = 0 \text{ V}$, $U_{O1} = -U_D = -0.2 \text{ V}$, $U_{O1} = +U_Z = \pm 5 \text{ V}$.

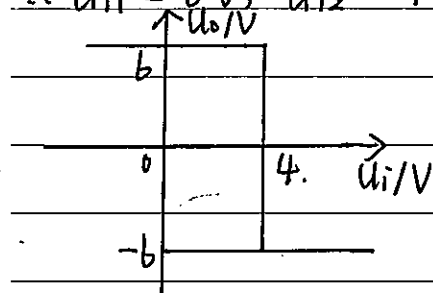


(c). $U_O = \pm U_T = \pm 6 \text{ V}$.

$$U_P = \frac{R_1}{R_1 + R_2} U_O + \frac{R_2}{R_1 + R_2} U_{REF}$$

令 $U_P = U_N = U_1$, 则 $U_N = \frac{1}{3} U_O + 2$.

$\therefore U_{T1} = 0 \text{ V}$, $U_{T2} = 4 \text{ V}$.



(d). $U_O = \pm U_Z = \pm 6 \text{ V}$.

$$U_P = \frac{R_2}{R_1 + R_2} U_1 + \frac{R_1}{R_1 + R_2} U_O = \frac{1}{3} U_O + \frac{2}{3} U_1$$

令 $U_P = U_N = 3 \text{ V}$, $U_1 = 9 = \frac{1}{2} (9 - U_O)$

$\therefore U_{T1} = 1.5 \text{ V}$, $U_{T2} = 7.5 \text{ V}$.

